



FIT5057 Project Management Assignment Two – Team Assignment

The Stellar Nexus– A Galactica Odyssey

September 2024

Student ID and Full Name: 34525416, Rishabh Ray

Team number: 1302

Applied class: Thursday, 10am-12pm



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References

GENERATIVE AI - Acknowledgement of use



DELIVERABLE 1: Project Integration Management

Task 1 Project Charter

Project Title: The Stellar Nexus – A Galactica Odyssey

Project Start Date: September 11, 2024

Projected Finish Date: June 15, 2027

Budget Information: Total project cost is AUD 1,071,850 which includes the 20% contingency

Project Manager:

Name: Rishabh Ray

Phone: +61 0481334132

e-mail: rray0009@student.monash.edu

Project Objectives:

The Stellar Nexus is created to address a gap in NexaForge's RPG portfolio by developing a Sci-Fi exploration game with non-linear storytelling and procedurally generated environments.

1. This game will expand NexaForge's player base and improve **Day 1 retention to 45%, Day 7 retention to 30%, and three-month retention to 16%.**
 2. This game will be contributing to the company's **17% revenue growth target over three years**
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Main Project Success Criteria:

1. Completion of game design documentation before Q1 2025.
 2. A playable prototype by Q4 2025.
 3. Game launch by Q1 2027.
 4. Achieving target retention metrics: 45% (Day 1), 30% (Day 7), and 16% (three months).
 5. Return on Investment (ROI) of 42.27% and positive NPV of AUD 767,297
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Project development approach:

The project will use a hybrid of Waterfall and Agile methodologies. *(Mahadevan, L., Kettinger, W. J., & Meservy, T. O. (2015))* The Waterfall model will be applied for pre-production and documentation, while Agile sprints will guide iterative game development with continuous feedback from NexaForge and alpha testers.

Roles and Responsibilities

Role in the project	Name	Position in the organisation/contract	Contact Information
Project Manager	Rishabh Ray	Project Manager	rray0009@student.monash.edu
NexaForge's CEO	John Bufort	Project Sponsor	john.bufort@nexaforge.com
Lead Developer	Darwin Gale	Gameplay & Mechanics Dev	darwingale@student.monash.edu
Art Director	Scully Ali	Art & UI/UX Design	ali.scully@student.monash.edu
QA Lead	Garwin Gupta	Quality Assurance Lead	gupta.garwin@student.monash.edu

Sign-off: (Signatures of all stakeholders listed)

Rishabh Ray

John Bufort

Darwin Gale

Scully Ali

Garwin Gupta

DELIVERABLE 2: Project Scope Management

Task 2.1 Requirements Traceability Matrix (RTM)

REQUIREMENTS TRACEABILITY MATRIX					
Project Name:	The Stellar Nexus– A Galactica Odyssey				
Project Manager Name:	Rishabh Ray				
Project Description:	The Stellar Nexus is created to address a gap in NexaForge's RPG portfolio by developing a Sci-Fi exploration game with non-linear storytelling and procedurally generated environments. This game will expand NexaForge's player base and improve Day 1 retention to 45%, Day 7 retention to 30%, and three-month retention to 16%. This game will be contributing to the company's 17% revenue growth target over three years.				
<i>ID</i>	<i>Requirements (Functional or Non-Functional)</i>	<i>Assumption(s) and/or Customer Need(s)</i>	<i>Category</i>	<i>Source</i>	<i>Status</i>
R01	Procedurally generated content for galaxies and planets	Needed for non-linear exploration	Functional	Design Team	In Progress
R02	Diplomacy and decision-making mechanics	Allows players to influence galactic balance	Functional	Design Team	Planned

R03	Non-linear storytelling with multiple outcomes	Critical for dynamic narrative	Functional	Concept Art	In Progress
R04	Hand-painted visual elements for characters and environments	Aesthetic appeal and artistic direction	Non-Functional	Art Team	In Progress
R05	Cross-platform compatibility for PC, consoles	Necessary for broader audience reach	Functional	Dev Team	Planned
R06	Procedurally generated player interactions	Diplomacy and decision-making mechanics	Functional	Dev Team	Planned
R07	In-game analytics for player tracking	Key for improving retention rates	Functional	Dev Team	Planned
R08	Post-launch updates and bug fixes	Supports long-term engagement	Non-Functional	Dev Team	Planned

Task 2.2 Project Scope Statement

PROJECT	DATE
The Stellar Nexus– A Galactica Odyssey	Project Start Date: September 11, 2024 Projected Finish Date: June 15, 2027
PROJECT BACKGROUND, OBJECTIVES and OUTCOMES	
The Stellar Nexus is created to address a gap in NexaForge's RPG portfolio by developing a Sci-Fi exploration game with non-linear storytelling and procedurally generated environments. This game will expand NexaForge's player base and improve Day 1 retention to 45%, Day 7 retention to 30%, and three-month retention to 16%. This game will be contributing to the company's 17% revenue growth target over three years. Objectives and Outcomes: 1. Develop a fully functional RPG game with procedural content, non-linear storytelling, and unique diplomacy mechanics. 2. Deliver a playable prototype and GDD. 3. Deliver post-launch support, including bug fixes and additional content	
HIGH-LEVEL PROJECT REQUIREMENTS, FUNCTIONAL & NON-FUNCTIONAL	
1. Functional: • Procedural generation system for dynamic exploration. • Diplomacy mechanics allowing players to influence game outcomes. • In-game analytics for player tracking 2. Non-Functional: • Artistic, hand-painted visuals. • Cross-platform compatibility for PC and consoles	

DELIVERABLES

1. Complete game with procedural generation, non-linear storytelling, and dynamic diplomacy.
2. Game design document (GDD) and a playable prototype.
3. Post-launch support for bug fixes and updates

ASSUMPTIONS

1. The game will be completed within 2 years timeframe with the post launch support for 2 months.
2. Development will follow a hybrid of Waterfall method and Agile methodology, allowing iterative improvements over 2.5 years.

PLANNED START AND END DATE

1. **Project Start Date:** September 11, 2024
2. **Projected Finish Date:** June 15, 2027

ESTIMATED COST

1. The estimated budget for the project is: AUD 1,071,850 which includes a 20% contingency cost

DELIVERABLE 3: Project Work Breakdown Structure (WBS) and Schedule

Task 3.1 Work Breakdown Structure (WBS)

Project Title: The Stellar Nexus

- 1.0 : Game Design and Development (Level 1)
 - 1.1 : Pre-Production (Level 2)
 - 1.1.1 : Concept Development (Level 3)
 - 1.1.1.1 : Create Concept Art for Characters and Environments (Level 4)
 - 1.1.1.2 : Determine Game Theme and Style (Level 4)
 - 1.1.1.3 : Create Storyline and Narrative (Level 4)

Completion of Concept Development

- 1.1.2 : Game Design Document (GDD) (Level 3)
 - 1.1.2.1 : Write and Finalize Game Design Document (Level 4)
 - 1.1.2.2 : Approve Game Mechanics, Story, and Features (Level 4)

Approval of Game Design Document

- 1.2 : Procedural Generation Design (Level 2)
- 1.3 : Non-Linear Storyline and Diplomacy Mechanics (Level 2)
- 2.0 : Production (Level 1)

- 2.1 : Core Game Development (Level 2)
 - 2.1.1 : Game Mechanics (Level 3)
 - 2.1.1.1 : Develop Player Movement and Combat Systems (Level 4)
 - 2.1.1.2 : Build Diplomacy and Decision-Making Mechanics (Level 4)
- 2.2 : Procedural Content Development (Level 2)
- 2.3 : Art Asset Creation (Characters, Environments, UI) (Level 2)

Core Game Development Complete

- 3.0 : Quality Assurance (Level 1)
 - 3.1 : Testing and Iteration (Level 2)
 - 3.1.1 : Internal and Alpha Testing (Level 3)
 - 3.1.1.1 : Internal Playtesting (Level 4)
 - 3.1.1.2 : Conduct Alpha Testing and Feedback Collection (Level 4)
 - 3.2 : Beta Testing (Level 2)
 - 3.3 : Bug Fixes and Gameplay Refinements (Level 2)
- 4.0 : Project Management (Level 1)
 - 4.1 : Schedule and Milestone Management (Level 2)
 - 4.1.1 : Timeline Management (Level 3)
 - 4.1.1.1 : Track Progress and Adjust Milestones (Level 4)
 - 4.2 : Resource Allocation and Budget Management (Level 2)
 - 4.3 : Risk Identification and Mitigation (Level 2)
- 5.0 : Post-Production (Level 1)
 - 5.1 : Bug Fixes and Final Tweaks (Level 2)

- 5.1.1 : Internal Testing Bug Fixes (Level 3)
 - 5.1.1.1 : Address Bugs Identified During Internal Testing (Level 4)
 - 5.1.1.2 : Optimize Game Performance (Level 4)
- 5.1.2 : Alpha and Beta Testing Feedback (Level 3)
 - 5.1.2.1 : Collect and Analyze Feedback from Alpha/Beta Testing (Level 4)
 - 5.1.2.2 : Implement Gameplay Improvements Based on Feedback (Level 4)
- 5.2 : Game Launch and Platform Distribution (Level 2)
 - 5.2.1 : Digital Distribution Setup (Level 3)
 - 5.2.1.1 : Upload to Online Platforms (PlayStation, Xbox, etc.) (Level 4)
 - 5.2.1.2 : Configure Game for Region-Specific Releases (Level 4)

Game Launch

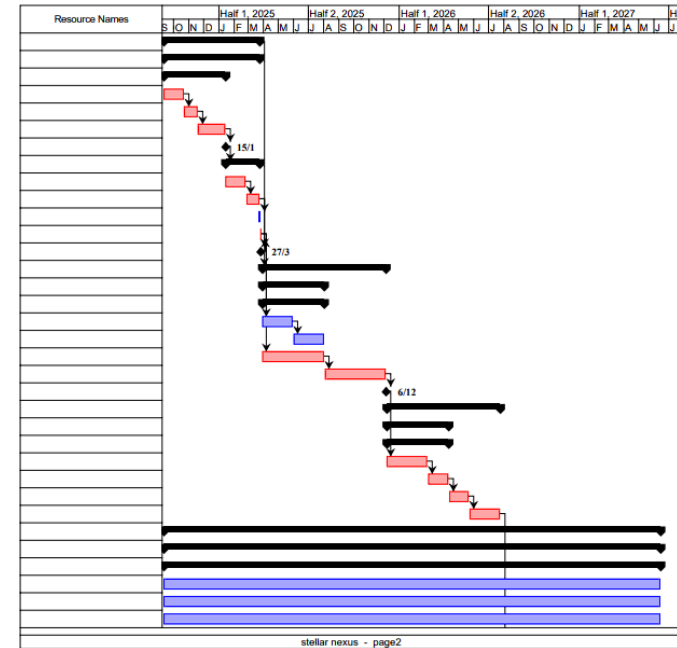
- 6.0 : Post-Launch Support (Level 1)
 - 6.1 : Post-Launch Bug Fixes (Level 2)
 - 6.1.1 : Monitor Bug Reports (Level 3)
 - 6.1.1.1 : Set Up Bug Tracking System for Players (Level 4)
 - 6.1.1.2 : Analyse and Prioritize Reported Bugs (Level 4)

Task 3.2 Team collaboration and Short reflection

While preparing the WBS and Gantt Chart along with my team, I came to the realization how multiple project managers can help with finding different flaws and loopholes within a project and create a better project plan for the entire team. While creating the WBS and Gantt Chart I understood the importance of the milestones and how every single phase of the project has to flow in a certain iterative manner to productively and timely complete the phases and make good use of the allocated resources and the team.

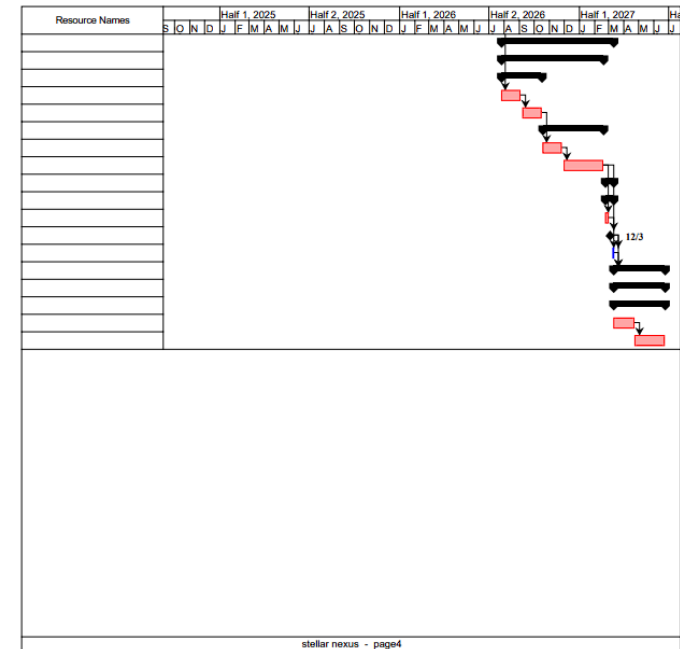
Task 3.3 Gantt Chart

ID	Name	Duration	Start	Finish	Predecessors
1	1.0 : Game Design and Development (Level 1)	140 days	11/9/24 8:00 AM	25/3/25 5:00 PM	
2	1.1 : Pre-Production (Level 2)	140 days	11/9/24 8:00 AM	25/3/25 5:00 PM	
3	1.1.1 : Concept Development (Level 3)	90 days	11/9/24 8:00 AM	14/1/25 5:00 PM	
4	1.1.1.1 : Create Concept Art for Characters and Envir...	30 days	11/9/24 8:00 AM	22/10/24 5:00 PM	
5	1.1.1.2 : Determine Game Theme and Style (Level 4)	20 days	23/10/24 8:00 AM	19/11/24 5:00 PM	4
6	1.1.1.3 : Create Storyline and Narrative (Level 4)	40 days	20/11/24 8:00 AM	14/1/25 5:00 PM	5
7	Completion of Concept Development	0 days	14/1/25 5:00 PM	14/1/25 5:00 PM	6
8	1.1.2 : Game Design Document (GDD) (Level 3)	50 days	15/1/25 8:00 AM	25/3/25 5:00 PM	7
9	1.1.2.1 : Write and Finalize Game Design Document (L...	30 days	15/1/25 8:00 AM	25/2/25 5:00 PM	
10	1.1.2.2 : Approve Game Mechanics, Story, and Featur...	20 days	26/2/25 8:00 AM	25/3/25 5:00 PM	9
11	Approval of Game Design Document	0 days	25/3/25 5:00 PM	25/3/25 5:00 PM	10
12	1.2 : Procedural Generation Design (Level 2)	2 days	27/3/25 8:00 AM	28/3/25 5:00 PM	13
13	1.3 : Non-Linear Storyline and Diplomacy Mechanics (Level 2)	1 day	26/3/25 8:00 AM	26/3/25 5:00 PM	10
14	2.0 : Production (Level 1)	180 days	31/3/25 8:00 AM	5/12/25 5:00 PM	1
15	2.1 : Core Game Development (Level 2)	90 days	31/3/25 8:00 AM	1/8/25 5:00 PM	
16	2.1.1 : Game Mechanics (Level 3)	90 days	31/3/25 8:00 AM	1/8/25 5:00 PM	
17	2.1.1.1 : Develop Player Movement and Combat Syste...	45 days	31/3/25 8:00 AM	30/5/25 5:00 PM	12
18	2.1.1.2 : Build Diplomacy and Decision-Making Mechan...	45 days	2/6/25 8:00 AM	1/8/25 5:00 PM	17
19	2.2 : Procedural Content Development (Level 2)	90 days	31/3/25 8:00 AM	1/8/25 5:00 PM	12
20	2.3 : Art Asset Creation (Characters, Environments, UI) (...)	90 days	4/8/25 8:00 AM	5/12/25 5:00 PM	19
21	Core Game Development Complete	0 days	5/12/25 5:00 PM	5/12/25 5:00 PM	20
22	3.0 : Quality Assurance (Level 1)	165 days	8/12/25 8:00 AM	24/7/26 5:00 PM	
23	3.1 : Testing and Iteration (Level 2)	90 days	8/12/25 8:00 AM	10/4/26 5:00 PM	
24	3.1.1 : Internal and Alpha Testing (Level 3)	90 days	8/12/25 8:00 AM	10/4/26 5:00 PM	
25	3.1.1.1 : Internal Playtesting (Level 4)	60 days	8/12/25 8:00 AM	27/2/26 5:00 PM	21
26	3.1.1.2 : Conduct Alpha Testing and Feedback Collecti...	30 days	2/3/26 8:00 AM	10/4/26 5:00 PM	25
27	3.2 : Beta Testing (Level 2)	30 days	13/4/26 8:00 AM	22/5/26 5:00 PM	26
28	3.3 : Bug Fixes and Gameplay Refinements (Level 2)	45 days	25/5/26 8:00 AM	24/7/26 5:00 PM	27
29	4.0 : Project Management (Level 1)	720 days	11/9/24 8:00 AM	15/6/27 5:00 PM	
30	4.1 : Schedule and Milestone Management (Level 2)	720 days	11/9/24 8:00 AM	15/6/27 5:00 PM	
31	4.1.1 : Timeline Management (Level 3)	720 days	11/9/24 8:00 AM	15/6/27 5:00 PM	
32	4.1.1.1 : Track Progress and Adjust Milestones (Level 4)	720 days	11/9/24 8:00 AM	15/6/27 5:00 PM	
33	4.2 : Resource Allocation and Budget Management (Level 2)	720 days	11/9/24 8:00 AM	15/6/27 5:00 PM	
34	4.3 : Risk Identification and Mitigation (Level 2)	720 days	11/9/24 8:00 AM	15/6/27 5:00 PM	



	🕒	Name	Duration	Start	Finish	Predecessors
35		5.0 : Post-Production (Level 1)	164 days	27/7/26 8:00 AM	11/3/27 5:00 PM	
36		5.1 : Bug Fixes and Final Tweaks (Level 2)	150 days	27/7/26 8:00 AM	19/2/27 5:00 PM	
37		5.1.1 : Internal Testing Bug Fixes (Level 3)	60 days	27/7/26 8:00 AM	16/10/26 5:00 PM	
38		5.1.1.1 : Address Bugs Identified During Internal Testi...	30 days	27/7/26 8:00 AM	4/9/26 5:00 PM	28
39		5.1.1.2 : Optimize Game Performance (Level 4)	30 days	7/9/26 8:00 AM	16/10/26 5:00 PM	38
40		5.1.2 : Alpha and Beta Testing Feedback (Level 3)	90 days	19/10/26 8:00 AM	19/2/27 5:00 PM	
41		5.1.2.1 : Collect and Analyze Feedback from Alpha/Be...	30 days	19/10/26 8:00 AM	27/11/26 5:00 PM	39
42		5.1.2.2 : Implement Gameplay Improvements Based o...	60 days	30/11/26 8:00 AM	19/2/27 5:00 PM	41
43		5.2 : Game Launch and Platform Distribution (Leve...	14 days	22/2/27 8:00 AM	11/3/27 5:00 PM	
44		5.3.1 : Digital Distribution Setup (Level 3)	14 days	22/2/27 8:00 AM	11/3/27 5:00 PM	
45		5.3.1.1 : Upload Game to Online Platforms (Steam, Pla...	7 days	22/2/27 8:00 AM	2/3/27 5:00 PM	42
46		5.3.1.2 : Configure Game for Region-Specific Releases...	7 days	3/3/27 8:00 AM	11/3/27 5:00 PM	45
47		Game Launch	0 days	11/3/27 5:00 PM	11/3/27 5:00 PM	42;46
48		6.0 : Post-Launch Support (Level 1)	75 days	12/3/27 8:00 AM	24/6/27 5:00 PM	46;47
49		6.1 : Post-Launch Bug Fixes (Level 2)	75 days	12/3/27 8:00 AM	24/6/27 5:00 PM	
50		6.1.1 : Monitor Bug Reports (Level 3)	75 days	12/3/27 8:00 AM	24/6/27 5:00 PM	
51		6.1.1.1 : Set Up Bug Tracking System for Players (Lev...	30 days	12/3/27 8:00 AM	22/4/27 5:00 PM	
52		6.1.1.2 : Analyze and Prioritize Reported Bugs (Level 4)	45 days	23/4/27 8:00 AM	24/6/27 5:00 PM	51

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Milestones:

- Completion of Concept Development
- Approval of Game Design Document
- Core Game Development Complete
- Game Launch

Rationale and Justification for the Gantt Chart

1. Project Management (Q4 2024 – Q1 2027)

- **Rationale:** Project management spans the entire project to ensure continuous oversight, resource allocation, milestone tracking, and risk management because managing contractors, coordinating between team members, and monitoring progress are necessary throughout the development lifecycle. *(Too, E. G., & Weaver, P. (2014))*

2. Game Design Document (GDD) (Q4 2024 – Q2 2025)

- **Rationale:** The GDD is foundational for the game's direction and mechanics, taking time to be fully developed, reviewed, and finalized. Continuous iterations are needed to refine game mechanics, story, and features, which justifies the extended timeline. *(Khan, T. H. (2021))*

3. Procedural Content Design (Q2 2025 – Q4 2025)

- **Rationale:** Procedural generation is a core feature of the game, and it requires extensive design to ensure the content dynamically adapts to player decisions. *(Beukman, M., Cleghorn, C. W., & James, S. (2022, July))* The long timeline allows for balancing and integration with other game mechanics like exploration and diplomacy.

4. Core Game Development (Q1 2025 – Q4 2026)

- **Rationale:** This task covers the full implementation of game mechanics, systems, and features. Given the game's complexity (procedural generation, non-linear storyline, diplomacy), development is spread over a long period to ensure thorough coding, integration, and continuous testing. *(Kourkoulou, T. (2023).)*

5. Testing and QA (Q3 2025 – Q2 2026)

- **Rationale:** Quality assurance requires multiple phases, starting early to catch issues during development. *(Politowski, C., Petrillo, F., & Guéhéneuc, Y. G. (2021, May))* Testing spans a long time to ensure bugs are addressed continuously, especially as different game systems are integrated and refined.

6. Post-Launch Support (Q4 2026 – Q1 2027)

- **Rationale:** Ongoing support after the game's release is critical for addressing player-reported bugs, delivering updates, and maintaining player engagement. This extended post-launch phase ensures stability and updates for the game after release. *(Josef, A., Van Lepp, A., & Carper, M. (2022))*

Conclusion:

These major tasks span longer periods due to the need for ongoing refinement, integration, and testing, as well as continuous project oversight. The Gantt chart reflects these long-term commitments to ensure the project's successful completion.

DELIVERABLE 4: Project Cost Management

Task 4.1 Cost Model:

WBS Items	Units/Hours	Cost/Unit or Hr	Subtotal	WBS Level 2 Total	% of Total
1. Project Management			\$327,100.00		30.52%
1.1 Project Manager	740	\$75.00	\$55,500.00		
1.2 Project Team Members (Designers, Devs)	1400	\$75.00	\$105,000.00		
1.3 Contractors (10% of development costs)			\$90,000.00		
1.4 Game Developers (2 developers)	700	\$42.00	\$29,400.00		
1.5 Data Scientist (1 data scientist)	420	\$60.00	\$25,200.00		
1.6 QA/Test Analyst (1 QA)	220	\$100.00	\$22,000.00		
2. Hardware			\$85,000.00		7.93%
2.1 Game Development PCs	15	\$3,500.00	\$52,500.00		
2.2 Servers	3	\$5,000.00	\$15,000.00		
2.3 Backup Devices	5	\$3,500.00	\$17,500.00		
3. Software			\$213,904.00		19.96%
3.1 Game Engine License	1	\$53,452.00	\$53,452.00		
3.2 Procedural Content Generation Software	1	\$55,452.00	\$55,452.00		
3.3 Software Tools (Art, Animation, Audio)	15	\$7,000.00	\$105,000.00		
4. Testing Cost			\$96,846.00		9.04%
5. Training and Support			\$149,000.00		13.90%
5.1 Game Design Team Training	100	\$500.00	\$50,000.00		
5.2 Travel and Conferences	10	\$1,500.00	\$15,000.00		
5.3 Post-Launch Support	1200	\$70.00	\$84,000.00		
6. Reserves			\$200,000.00		18.66%
Total Project Cost Estimate			\$1,071,850.00		100%

Task 4.2 Cost Baseline

WBS Items	Quarter 4, 2024	Quarter 1, 2025	Quarter 2, 2025	Quarter 3, 2025	Quarter 4, 2025	Quarter 1, 2026	Quarter 2, 2026	Quarter 3, 2026	Quarter 4, 2026	Quarter 1, 2027	Total
1. Project Management											\$327,100.00
1.1 Project Manager	\$5,550.00	\$5,550.00	\$5,550.00	\$5,550.00	\$5,550.00	\$5,550.00	\$5,550.00	\$5,550.00	\$5,550.00	\$5,550.00	\$55,500.00
1.2 Project Team Members (Designers, Devs)	\$10,500.00	\$10,500.00	\$10,500.00	\$10,500.00	\$10,500.00	\$10,500.00	\$10,500.00	\$10,500.00	\$10,500.00	\$10,500.00	\$105,000.00
1.3 Contractors (10% of development costs)			\$22,500.00	\$22,500.00	\$22,500.00	\$22,500.00					\$90,000.00
1.4 Game Developers (2 developers)	\$2,940.00	\$2,940.00	\$2,940.00	\$2,940.00	\$2,940.00	\$2,940.00	\$2,940.00	\$2,940.00	\$2,940.00	\$2,940.00	\$29,400.00
1.5 Data Scientist (1 data scientist)	\$6,300.00		\$6,300.00		\$6,300.00		\$6,300.00				\$25,200.00
1.6 QA/Test Analyst (1 QA)				\$4,400.00		\$4,400.00		\$4,400.00	\$4,400.00	\$4,400.00	\$22,000.00
2. Hardware											\$85,000.00
2.1 Game Development PCs	\$15,000.00	\$37,500.00									\$52,500.00
2.2 Servers		\$15,000.00									\$15,000.00
2.3 Backup Devices		\$17,500.00									\$17,500.00
3. Software											\$213,904.00
3.1 Game Engine License		\$53,452.00									\$53,452.00
3.2 Procedural Content Generation Software			\$55,452.00								\$55,452.00
3.3 Software Tools (Art, Animation, Audio)		\$35,000.00	\$35,000.00	\$35,000.00							\$105,000.00
4. Testing (10% of hardware and software)				\$24,211.50	\$24,211.50	\$24,211.50			\$24,211.50		\$96,846.00
5. Training and Support											\$149,000.00
5.1 Game Design Team Training		\$25,000.00	\$25,000.00								\$50,000.00
5.2 Travel and Conferences	\$5,000.00	\$5,000.00		\$5,000.00							\$15,000.00
5.3 Post-Launch Support										\$84,000.00	\$84,000.00
6. Reserves	\$40,000.00				\$40,000.00				\$40,000.00	\$80,000.00	\$200,000.00
Total Project Cost	\$85,290.00	\$207,442.00	\$163,242.00	\$110,101.50	\$112,001.50	\$70,101.50	\$25,290.00	\$23,390.00	\$87,601.50	\$187,390.00	\$1,071,850.00

("HAYS IT Contractor Rates Guide FY23/34," 2023/2024))

(Unreal Engine. (2023))

(Talent.com: Job Search | Find Jobs Hiring Near You. (2024))

(Newegg. (2016))

(Lead, Y. P., Product Marketing. (2023))

(JB Hi-Fi. (2023))

(How Much Does It Cost to Develop a Game. (2022, May 6))

(Company Salaries. (2024))

Task 4.3 Assumptions

1. Project Management

- **1.1 Project Manager:** Costs evenly spread from Q4 2024 to Q1 2027, ensuring continuous oversight throughout the project.
- **1.2 Project Team Members (Designers, Devs):** Allocated evenly from Q4 2024 to Q1 2027 to support ongoing design and development work.

2. Contractors

- **1.3 Contractors (10% of development costs):** Engaged during Q3 2025 to Q2 2026, providing additional expertise during intensive development and testing.

3. Game Development

- **1.4 Game Developers:** Costs spread from Q4 2024 to Q1 2027, reflecting their continued role in core development.
- **1.5 Data Scientist:** Engaged in specific quarters (Q4 2024, Q2 2025, and Q4 2025) to support data analytics and procedural content generation.
- **1.6 QA/Test Analyst:** Focused on testing from Q3 2025 to Q4 2026, ensuring quality and bug fixes during critical development phases.

4. Hardware & Software

- PCs, servers, and backup devices acquired in Q4 2024 and Q1 2025 for early development. Software tools and licenses purchased between Q1 and Q2 2025 to support game design and development.

5. Testing

- Incremental Approach starting in Q3 2025 to ensure early feedback and bug fixes, ramping up through Q4 2026.

6. Training and Support

- Training conducted in Q1 and Q2 2025, with Post-Launch Support from Q4 2026 to Q1 2027 to handle updates and player engagement.

7. Reserves

- Allocated in Q4 2024, Q4 2025, and Q1 2027 for risk management and unforeseen expenses.

DELIVERABLE 5: Project Risk Management

Task 5.1 Risk Register

RISK ID	RANK	RISK DESCRIPTION	IMPACT DESCRIPTION	IMPACT LEVEL	PROBABILITY LEVEL	PRIORITY LEVEL	RISK RESPONSE	RISK OWNER
R01	1	Complexity in procedural generation algorithm	May affect gameplay variety and player retention	High	Medium	3	Keep doing iterative testing	Lead Dev
R02	2	Competition with similar RPG titles	Risk of reduced market share	High	High	3	Do better marketing	Marketing
R03	3	Delays in creation of art asset	Could push back the overall project timeline	High	Medium	3	Plan buffer in the timeline	Art Director
R04	4	Early completion of the cross-platform integration due to better emulation systems to convert the prepared game	May cause misalignment in the project phases timeline due to asynchronous development	Medium	Medium	2	Reallocation of resources to complete the other phases according to the plan	Project Manager

Task 5.2 Probability & Impact Rationale

Each risk has been evaluated based on its potential impact on the project and the likelihood of occurrence, allowing for a structured approach to risk management.

- **R01: Complexity in Procedural Generation Algorithm**

This risk is rated with a **High Impact** level due to its critical role in gameplay variety and player retention. Although it has a **Medium Probability** of occurring, to avoid its potential to adversely affect the game experience we will be doing rigorous testing and mitigation strategies. *(Gellel, A., & Sweetser, P. (2020, September))*

- **R02: Competition with Similar RPG Titles**

This risk presents a **High Impact** due to the potential loss of market share in a competitive landscape. It is also assessed with a **High Probability** level, for the same we will be focusing on marketing strategies to differentiate the game and attract more players.

- **R03: Delays in Art Asset Creation**

The impact of this risk is **High**, as it could significantly delay the overall project timeline. The probability is assessed as **Medium**, we can ensure that any delays do not derail the project by taking buffer time for the development. *(Biffel, S., Musil, J., Musil, A., Meixner, K., Lüder, A., Rinker, F., ... & Winkler, D. (2021, September))*

- **R04: Early Completion of Cross-Platform Integration**

While this risk has a **Medium Impact** level, its probability is also **Medium**. Early completion will lead to misalignment of the project phases, so we will be reallocating resources to stay on track with the overall project plan. *(Blanco, J. Z., & Lucrédio, D. (2021))*

Task 5.3 The Matrix & Analysis

The Probability/Impact Matrix serves as a vital tool in visualizing and prioritizing project risks based on their potential effects and likelihood of occurrence. The matrix categorizes risks into four quadrants:

- **High Probability / High Impact:** Risks such as the complexity of the procedural generation algorithm (R01) and competition with similar RPG titles (R02) fall into this category. These risks will require immediate attention and robust mitigation strategies. We will be making efforts into iterative testing for the algorithm and a strong marketing campaign to address competition.
- **Medium Probability / High Impact:** Delays in art asset creation (R03) will go here, for this we will be allocating buffer time to manage this risk. Monitoring progress in this area will help mitigate the potential impact on the overall timeline.
- **Medium Probability / Medium Impact:** The risk associated with early completion of cross-platform integration (R04) is considered lower priority. It will require monitoring, the strategy of reallocating resources allows for flexibility without derailing project phases.

PROBABILITY	5					
	4				R01 R02	
	3			R04	R03	
	2					
	1					
		1	2	3	4	5
		IMPACT				

DELIVERABLE 6: Project Quality Management

Task 6.1 Quality Standards/Requirements

Quality Standard 1: Frame Rate more than 60fps for 5% of the user

Justification: Higher frame rates (e.g., 120 and 180fps) can ensure better user performance than lower rates. Interestingly, we also found that at 60fps and when users are faced with fast-moving objects, they tend to adopt a strategy to compensate for the lack of visual details by predicting or filling the gaps to try to meet the performance needs. *(Wang, J., Shi, R., Zheng, W., Xie, W., Kao, D., & Liang, H. N. (2023))*

Quality Standard 2: User Satisfaction remains above 3 star 80% of the user for User

Justification: A user satisfaction rate of above 3 star is good for our game as in the initial years getting higher reviews is tough for any game and a 3 star average rating is considered to be a success in the initial years for any RPG game, with iterative updates the ratings can be improved. *(Lin, D., Bezemer, C. P., Zou, Y., & Hassan, A. E. (2019), Achterbosch, L., Pierce, R., & Simmons, G. (2008).)*

Quality Standard 3: Bug Resolution Efficiency should be 75% before launch

Justification: The game needs to be bug free before the launch and a rate of 75% bug riddance is considered good for a new release of the game. It will be done by increasing testing and using the contractors to help with debugging. *(Zhang, I. (2016))*

Quality Standard 4: Diplomacy System Usage should be used by 50% players

Justification: The use of diplomacy feature would suggest that the system is working as intended and help innovatively get the feature even better over time. A 50% metric has been taken so that the user base is wide enough for the testing of the platform. The main goal is to provide players with freedom similar to Diplomacy *(Valach, M. (2024))*

Why these standards are right for the project:

The success of *The Stellar Nexus* heavily depends on player satisfaction, smooth performance across multiple platforms, and a bug-free experience, as these directly impact player retention and overall engagement. Since NexaForge's target is a broader audience with long-term engagement, achieving high performance and satisfaction metrics is critical for sustaining interest and ensuring ROI.

Assumptions:

1. **Hardware Assumptions:** It is assumed that the game will be tested on mid-range to high-end gaming systems. Performance standards are based on hardware configurations commonly available in the market during the release period.
2. **Testing Limitations:** The performance metrics assume that testing will cover 90% of all potential use cases, as it's impractical to account for every possible user setup.
3. **Content Update Pace:** The quality of the game post-launch will depend on periodic updates, assuming that the game's architecture supports modular content deployment for future improvements

Task 6.2 Metrics and Measurement

Metric	Measurement Method	Trigger for Action
Frame Rate	Measure FPS using in-game diagnostics during peak times.	FPS drops below 60 for 5% of users, trigger optimization efforts.
User Satisfaction	User surveys during beta testing and post-launch.	If satisfaction rates fall below 80%, conduct interviews for further feedback.
Bug Resolution Efficiency	Monitor bug closure rates during Alpha and Beta testing.	If less than 75% of critical bugs are resolved pre-launch, assign more resources to testing.
Diplomacy System Usage	Track how often players use diplomacy features.	If fewer than 50% of players engage with diplomacy, rework tutorial and mechanics for clarity.
Retention Metrics (Day 1/7/30)	Track retention rates with backend analytics for user engagement.	If retention metrics are below targets (45% Day 1, 30% Day 7), adjust content delivery.
Art Asset Feedback	Collect player feedback on game visuals during testing.	If more than 20% of feedback mentions visual dissatisfaction, iterate on art assets.

DELIVERABLE 7: Project Stakeholder and Communication Management

Task 7.1 Building Your Stakeholder Register

TITLE	ROLE IN PROJECT	CATEGORY	POWER LEVEL	INTEREST LEVEL	COMMS REQUIREMENTS	COMMS FREQUENCY	CONTACT
Project Manager	Project Manager	Internal	High	High	Email, Meetings	Weekly	rray0009@student.monash.edu
Lead Designer	Responsible for concept development	Internal	Medium	High	Email, Workshops	Bi-Weekly	garwin@sruent.monash.edu
Lead Developer	Oversees core game mechanics development	Internal	High	High	Email, Daily Standups	Daily	darwingale@student.monash.edu
Art Director	Oversees creation of characters, assets	Internal	Medium	High	Email, Workshops	Weekly	ali.scully@student.monash.edu
QA Lead	Oversees testing and iteration	Internal	Medium	Medium	Email, Reports	Bi-Weekly	trevor@student.monash.edu
NexaForge CEO	Project Sponsor	External	High	Medium	Reports, Presentations	Monthly	john.bufort@nexaforge.com
Marketing Manager	Responsible for market strategy and outreach	Internal	High	High	Email, Presentations	Weekly	kermin@student.monash.edu
External Testers	Provide feedback during Alpha and Beta Testing	External	Low	High	Email, Testing Reports	As Needed	cristine@student.monash.edu

Task 7.2 Engagement Strategy

1. **Stakeholder 1:** NexaForge CEO – John Bufort – Being the CEO of the company NexaForge we need to engage through monthly updates and milestone reports. Focus on high-level progress and ROI projections.
2. **Stakeholder 2:** QA Lead – Weekly status reports on testing results and bug fixes to ensure quality control remains aligned with project timelines, being an external testers there will be no biasness towards the results and the testing reports generated by Christine which will help us to create a better plan of working against the bugs.

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GENERATIVE AI - Acknowledgement of use

- AI & Generative AI tools has been SELECTIVELY used within this assessment to make the project plan close to realistic plans and to help create a working structure to distribute the cost of the project along with the help of the self-prepared Gantt chart and WBS.
- ChatGPT 4.0 has been selective used to rephrase the language of the document to sound more professional and in line with the requirements of Nexaforge Studio.

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